

# Staffing Responses to Nursing Home Ownership Changes: Evidence from U.S. Nursing Homes

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- facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024

# Identifying Ownership Changes: Why Ownership Is Non-Trivial

- Ownership is often **layered** (organizations owning organizations)
- We identify ownership changes at the **highest level of ownership present**

| Ownership Level           | Entity Type | Owner Name                    | Share |
|---------------------------|-------------|-------------------------------|-------|
| <i>Direct ownership</i>   |             |                               |       |
| Organization              | LLC         | BREDLEGS HOLDINGS LLC         | 25%   |
| Organization              | LLC         | ROCK HILL ACQUISITION III LLC | 75%   |
| <i>Indirect ownership</i> |             |                               |       |
| Individual                | Person      | LALLY, THOMAS                 | 25%   |
| Individual                | Person      | LITT, ALAN                    | 25%   |
| Individual                | Person      | LITT, JONATHAN                | 25%   |
| Individual                | Person      | STURM, JOSHUA                 | 25%   |

# Identifying Ownership Changes: Example (Legal Restructuring)

- HCRIS ownership change flags may reflect **purely legal restructuring**
- NHC ownership data allows us to verify whether economic control changes
- **Example from the data:**
  - ▶ *Before:*
    - ★ ESTES JAMES (84%)
    - ★ BURCHFIELD JOHN (10%)
    - ★ LEE CLAUDE (6%)
  - ▶ *After:*
    - ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
    - ★ JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
- We do **not** treat this as a meaningful ownership change

# Identifying Ownership Changes: Definition and Sample

- **Step 1 (flag):** HCRIS ownership change flag identifies candidate change events
- **Step 2 (validate & date):** NHC ownership panel identifies whether control changes and when
- **Ownership change criteria (NHC):**
  - ▶ An ownership change occurs between  $t - 1$  and  $t$  when **majority control (50%)** transfers
  - ▶ Partial / missing shares are normalized; missing ownership shares are assigned equal weights within a level
  - ▶ We do not flag changes driven by name changes or within-family reclassification (e.g., trust/LLC relabeling)
- **Final sample restriction):**
  - ▶ Facilities with **(0,0)** changes in both datasets, or **(1,1)** with dates within 6 months
  - ▶ Change month is the first month NHC indicates a change



# Ownership Change Agreement: NHC vs HCRIS

Cross-tab of Ownership Changes: NHC vs HCRIS

|                         |    |                           |       |     |
|-------------------------|----|---------------------------|-------|-----|
| Count of changes in NHC | 2+ | 1,137                     | 1,377 | 535 |
|                         | 1  | 1,874                     | 1,589 | 101 |
|                         | 0  | 6,217                     | 499   | 15  |
|                         |    | 0                         | 1     | 2+  |
|                         |    | Count of changes in HCRIS |       |     |

# Dependent Variables: Nurse Staffing (PBJ)

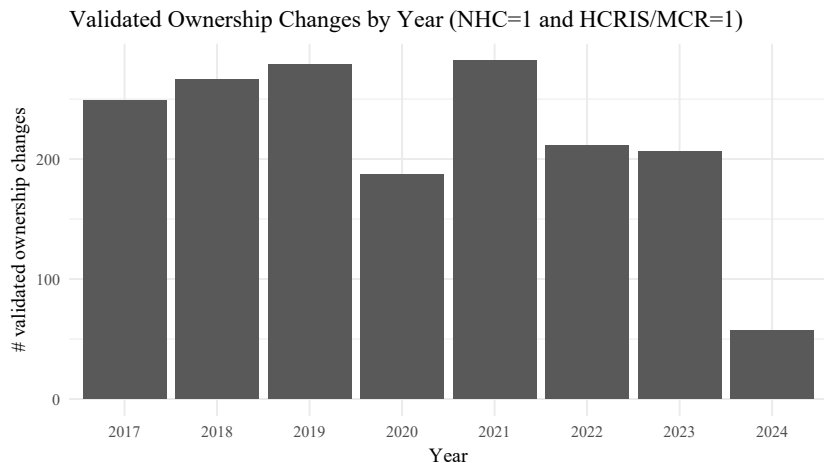
- We study four staffing measures:
  - ▶ **Registered Nurses (RN)**: clinical assessment, medication administration, supervision
  - ▶ **Licensed Practical Nurses (LPN)**: clinical support, medication-related tasks
  - ▶ **Certified Nursing Assistants (CNA)**: direct daily care (bathing, feeding, mobility)
  - ▶ **Total nursing staff**: RN + LPN + CNA
- Outcome is **hours per patient day (HPPD)**: staffing intensity standardized by resident census

$$\text{HPPD}_{s,i,t} = \frac{H_{s,i,t}}{\text{PatientDays}_{i,t}}$$

# PBJ Cleaning and Final Sample

- We apply CMS technical guidance to remove implausible staffing reports:
  - ① RN HPPD = 0 **and** LPN HPPD = 0
  - ② Total HPPD < 1.5 or > 12
  - ③ CNA HPPD > 5.25
- Additional restrictions:
  - ▶ Drop facilities with beds < 15
  - ▶ Drop hospital-based facilities
- **Final sample:**
  - ▶ 7,806 facilities
  - ▶ 632,231 facility-month observations
  - ▶ 1,589 facilities undergo a change of ownership, about 20% of the final sample

# Validated Ownership Changes Over Time (Yearly)



# Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD event study around ownership transitions
- Identification comes from **within-facility changes** relative to facilities not yet treated

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_{\tau} \mathbf{1}\{event\_time_{it} = \tau\} \cdot Treat_i + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}. \quad (1)$$

- $Y_{it}$ : staffing outcome measured in **HPPD**
- $\alpha_i$  facility FE,  $\lambda_t$  month FE;  $X'_{it}$  controls (beds, occupancy, payer mix, ownership type, chain, acuity)
- Standard errors clustered at the facility level

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- **Primary threat: anticipation / timing mismatch**
  - ▶ Staffing may adjust before the recorded transition month.



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- Administrative “event dates” may reflect reporting/legal dates, not the first operational change
- **Implication:** months immediately before  $\tau = 0$  may violate parallel trends

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- Post change coefficients compare to a pre-period less likely to be influenced by transition activity

# Assessing Parallel Trends

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- **Takeaway:**

- ▶ With the full pre-window (including  $\tau = -3, -2, -1$ ), pre-trends show evidence of violation.
- ▶ Excluding anticipatory months (donut) yields pre-trends more consistent with parallel trends.

# Parallel Trends Tests (Joint Wald) — Total Staffing

|                          | Total (Levels)      | Total (Logs)        |
|--------------------------|---------------------|---------------------|
| 2 Year Full Pre-Window   | 51.22 (23) [0.0006] | 49.22 (23) [0.0012] |
| 2 Year Window with Donut | 24.92 (20) [0.2046] | 22.26 (20) [0.3268] |
| 1 Year Window with Donut | 6.08 (8) [0.6387]   | 6.33 (8) [0.6102]   |

*Notes:* Joint Wald  $\chi^2$  test of  $H_0 : \beta_\tau = 0$  for all pre-treatment coefficients. Cells report  $\chi^2$  (df) [p-value].

Donut excludes  $\tau \in \{-3, -2, -1\}$ ; reference period is  $\tau = -4$ .

Full outcome-by-staff-type table available in appendix.

# Summary Effect: TWFE Difference-in-Differences

$$Y_{it} = \beta Post_{it} + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- $Post_{it} = 0$  before an ownership change and  $= 1$  in all months after a change
- $\beta$  summarizes the average post-change staffing effect
- Quantifies average changes for cleaner and relevant results